

ATmega328P 8-Bit AVR Microcontroller

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1. Architecture Overview & Operating Characteristics

The ATmega328P is a low-power CMOS 8-bit microcontroller based on the AVR enhanced RISC architecture. Executes 1 instruction per clock cycle, achieving throughputs approaching 1 MIPS per MHz.

Parameter	Operating Range	Critical Engineering Constraint
Operating Voltage	1.8V to 5.5V DC	5.0V required for 16 MHz / 20 MHz clock
Flash Program Memory	32 KB (with 0.5 KB Bootloader)	10,000 write/erase endurance cycles
SRAM Data Memory	2 KB internal	Stack overflow occurs if RAM fills
Max DC Current per I/O	40.0 mA absolute max	Recommended <= 20.0 mA per pin
Max DC Current VCC/GND	200.0 mA total device limit	Sum of all pin currents must not exceed 200mA

2. Primary Pinouts & Bus Interfaces (DIP-28 Package)

- Pin 1 (PC6 / RESET): Active-low reset. Must have external 10 kOhm pullup resistor to VCC. Pull to GND to reset MCU.
- Pin 7 (VCC) & Pin 8 (GND): Main digital power supply. Requires 0.1 uF decoupling capacitor directly adjacent.
- Pin 9 (XTAL1) & Pin 10 (XTAL2): Inverting oscillator amplifier pins. Connect 16 MHz crystal with twin 22 pF ceramic caps to GND.
- Pin 20 (AVCC): Power supply for Port A and Analog-to-Digital Converter. Must connect to VCC via low-pass LC filter (10 uH + 100 nF).
- Pin 21 (AREF): Analog reference voltage pin. Connect 100 nF capacitor to GND. Never tie directly to external 5V if internal 1.1V ref is active!
- Pin 27 (PC4 / SDA) & Pin 28 (PC5 / SCL): Hardware I2C bus. Requires external 4.7 kOhm pullup resistors on both lines.
- Pin 17 (MOSI), Pin 18 (MISO), Pin 19 (SCK): Hardware SPI bus used for high-speed communication and ICSP programming.

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3. Analog-to-Digital Converter (ADC) Guidelines

The ATmega328P includes a 10-bit successive approximation ADC with 6 multiplexed channels (Port C, ADC0 to ADC5).

ADC Clock Requirement: The ADC requires an input clock frequency between 50 kHz and 200 kHz for maximum 10-bit resolution. With a 16 MHz system clock, configure the ADC prescaler to 128 ($16 \text{ MHz} / 128 = 125 \text{ kHz}$).

Input Impedance: The ADC is optimized for analog signals with an output impedance of 10 kOhm or less. If source impedance is high, add a 10 nF capacitor to ground or buffer with an op-amp.

4. Troubleshooting Common Hardware Faults

- Fault 1: AVR programmer reports 'Device signature = 0x000000' or 'Target not responding'. Causes: Crystal clock is not oscillating (missing 22 pF caps), RESET pin is not pulled HIGH, or SPI MOSI/MISO connections are swapped. Solution: Verify 5V on Pin 7, verify 10 kOhm on Pin 1, check crystal.
- Fault 2: Microcontroller resets spontaneously when relay or motor turns on. Cause: Inductive voltage dip on VCC triggering internal Brown-Out Detector (BOD). Solution: Add a 470 uF electrolytic capacitor across the main power rails and install a flyback diode on the relay coil.
- Fault 3: ADC readings fluctuate wildly. Cause: AVCC (Pin 20) left disconnected or AREF (Pin 21) floating without decoupling cap. Solution: Connect AVCC to VCC and decouple AREF with 100 nF.